MILITARY
SPECIFICATION

MIL-I-10536A(SigC)
4 June 1953
Superseding
MIL-I-10536(SigC)
20 September 1950

INTERROGATOR SETS, COMPONENTS FOR

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1. SCOPE

1.1 Scope.- This specification covers the following components which are used with various interrogator sets. (See 6.1.)

	Ref. Par.
Receiver-Transmitter, Radio RT-211( )/TPX	3.9
Coder-Control, Interrogator Set KY-97( )/TPX	3.10
Coder-Control, Interrogator Set KY-99( )/UPX	3.11
Case, Receiver-Transmitter CY-944( )/TPX	3.8.2
Case, Standardized Components, Electrical CY-951( )/TPX	3.8.3
Case, Receiver-Transmitter CY-1324( )/UPX-6	3.8.4

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591	Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment).
NN-B-601	Boxes, Wood-Cleated-Plywood (For Domestic Shipment).
UU-T-111	Tape; Paper, Gummed (Kraft).
NN-B-631	Boxes, Wood, Wirebound (For Domestic Shipment).
LLL-B-631	Boxes, Fiber Corrugated (For Domestic Shipment).

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JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes; Wood, Cleated, Plywood.
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
MIL-B-107	Boxes, Wood, Wirebound (Overseas Type).
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116	Preservation, Methods of.
JAN-P-117	Packaging and Packing for Overseas Shipment - Bags, Interior Packaging.
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant.
MIL-B-131	Barrier-Material; Water-Vaporproof, Flexible.
JAN-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment: General Process for.
MIL-S-901	Shockproof-Equipment Class HI (High Impact); and Tests for (Shipboard Use).
MIL-C-3098	Crystal Units, Quartz.
MIL-P-11268	Parts, Materials, and Processes used in Signal Corps Equipment; General Specification.

U. S. ARMY

19-10	Filler, Interior Packing, Fiberboard, Corrugated, Single and Double Faced, for General Packing Purposes.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking of Shipments.

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DRAWINGS

SIGNAL CORPS

460-11	Group B-1 Sampling Plans.
460-12	Group B-2 Sampling Plans.
460-13	Group B-3 Sampling Plans.
460-14	Group B-4 Sampling Plans.
SC-D-15914	Moisture Resistance Test Cycle for Ground Signal Equipment.
SC-D-19999	Standard Extreme Temperature Cycle.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- The components covered by this specification are used as required with several interrogator sets.

3.2 Model(s).- The referenced model(s) shall be understood as the one(s) which, in accordance with the invitation for bids, will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein, the equipment shall conform to the following:

3.2.1 Construction.- Construction of the equipment shall conform to the model(s).

3.2.2 Functions.- The equipment shall incorporate all functions provided by the model(s).

3.2.3 Performance.- Performance of the equipment shall be at least equal to that of the model(s). When any performance characteristic of the model(s) is not specified herein, the following shall apply: As soon as the model(s) are received by the contractor, he shall adjust for optimum performance and prepare to conduct all measurements necessary to determine the performance characteristics which are not specified herein. The contracting officer shall be notified of the date when the measurements are to be taken, sufficiently in advance to send a representative to witness them. Measurements shall be taken by the contractor and three record copies thereof furnished to the contracting officer.

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3.3 Preproduction samples.-

3.3.1 Quantities.- After award of contract, in addition to the samples listed in Specification MIL-P-11268, two complete equipment shall be furnished for performance testing except Coder-Control, Interrogator Set KY-99( )/UPX for which 3 equipments shall be furnished; two for performance and one for bounce test of 4.3.3.

3.3.2 Compliance with specified requirements.-

3.3.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.3.2.3 for exception.)

3.3.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.3.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.3.2.3 When deviation from 3.3.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.3.3 Test data.- The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.3.4 Preproduction inspection.- Preproduction samples may be submitted by the Government to the group A inspection and nondestructive group B and group C inspection specified in table IV, and to the preproduction inspection specified in table I. Unlisted non-destructive inspection on preproduction samples of equipment and unlisted destructive or non-destructive inspection on separate samples of parts or materials may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, bench-handling, (4) shock, ballistic, (5) shock, drop, and (6) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

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Table I - Preproduction inspection

Inspection	Requirement Paragraph	Test Paragraph
Bounce	3.5.3.2	4.3.1
Bounce	3.5.3.2	4.3.2
Bounce	3.5.3.2	4.3.3
Shock, ballistic	3.5.3.2	4.6
Immersion	3.5.2	4.7.1
Moisture resistance	3.5	4.9

3.3.5 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.4 Parts, materials, and processes; general.- Parts, materials, and processes shall conform to Specification MIL-P-11268, unless otherwise specified herein.

3.4.1 Circuit labels.- Labels (schematics, tube layouts, etc.) of the type supplied on model in accordance with applicable requirements of Specification MIL-P-11268 shall be furnished for the following components:

Receiver-Transmitter Radio RT-211( )/TPX
Coder-Control, Interrogator Set KY-97( )/TPX
Coder-Control, Interrogator Set KY-99( )/UPX

3.4.2 Tropicalization.- In addition to tropicalization specified in Specification MIL-P-11268, the assembled equipment shall be treated in accordance with Specification JAN-T-152.

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3.5 Service conditions.- The equipment shall meet the following service conditions:

3.5.1 Equipment operating.- The equipment shall meet the requirements of this specification under any probable combination of the following:

a. Operation. Continuous operation in any specified manner, for a period of 1 year, alternating 23 hours on with 1 hour off.

b. Temperature. Any ambient temperature in the range of -40 to 131° F., including sun load.

c. Elevation. Any elevation up to 10,000 feet above sea level.

d. Relative humidity. Any relative humidity up to 95 percent.

e. Orientation. Any orientation up to 20 degrees from the normal operating orientation.

f. Dust. Dust as encountered during heavy traffic over dry clay roads.

3.5.2 Equipment non-operating.- The equipment shall meet the requirements of 3.5.1 after subjection to any one or more of the following non-operating conditions:

a. Temperature. Continuous exposure for 72 hours at 131° F. and 240 hours at -80° F., at any orientation.

b. Elevation. Elevations up to 50,000 feet above sea level.

c. Relative humidity. Relative humidity up to 100 percent, including condensation caused by temperature changes. (See 4.9.) Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.5.1.

d. Immersion. See 4.7.

3.5.3 Vibration, bounce, and shock.-

3.5.3.1 Vibration.- Except for internal resonance of parts and subassemblies specified herein, the equipment shall have no mechanical resonance below 55 cycles per second when shock mounts (if any) are blocked or removed.

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3.5.3.2 Bounce and shock.- The equipment shall meet the inspection of table II.

Table II - Bounce and shock inspection

Inspection	Inspection Paragraph	Requirements after Inspection
Bounce	4.3.1 4.3.2 4.3.3	Full specification performance. No physical damage except surface abrasion.
Shock test; ballistic	4.6	Components shall be operable. Minor mechanical damage is permissible. (Note A)
Shock test; bench-handling	4.4	Full specification performance. No physical damage.
Shock test; drop	4.5.1 4.5.2	Equipment shall be operable. Minor mechanical damage is permissible. (Note A)
NOTE A: After this shock test, equipment performance shall conform to the following: The equipment shall meet full specification performance.		

3.6 Facsimiles.- In addition to any facsimiles prescribed by subsidiary specifications, two each facsimile copies of the circuit labels (3.4.1) shall be submitted for approval to the contracting officer.

3.7 Definitions.-

3.7.1 Pulse rise time.- Rise time of a pulse is the time required for the leading edge of a pulse to rise from 10 to 90 percent of the maximum voltage amplitude.

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3.7.2 Pulse decay time.- Decay time of a pulse is the time required for the trailing edge of a pulse to decay from 90 to 10 percent of the maximum voltage amplitude.

3.7.3 Pulse "slope".- The term "slope" as applied to a pulse is the difference between the amplitude of the pulse at the instant when its leading edge passes from substantially linear slope to substantially non-linear slope, and its amplitude at the instant when the trailing edge of the pulse passes from substantially non-linear slope to substantially linear slope. Slope shall be expressed in terms of percent of maximum amplitude.

3.7.4 Pulse top.- The instantaneous amplitude of each pulse shall not at any instant between the point of the leading edge (as defined in 3.7.1) which is 90 percent of the maximum voltage amplitude of the pulse, to the point on the trailing edge (as defined in 3.7.2) which is 90 percent of the maximum voltage amplitude of the pulse, fall below a line which is 80 percent of the maximum voltage amplitude of the pulse and is parallel to the base line.

3.7.5 Pulse duration.- Pulse duration shall be measured between the points along the leading and trailing edges of a pulse which are at 50 percent of the maximum amplitude of the pulse.

3.7.6 Time measurements with pulse as reference.- Measurement of spacing between pulses, and other time measurements made with a pulse as reference, shall be made with reference to the instant when the leading edge of the pulse reaches 50 percent maximum amplitude.

3.8 Cases.- The cases shall be constructed in accordance with model and the following:

3.8.1 Waterproof construction.- The cases with contents shall be waterproof with all covers secured and shall be capable of being floated ashore during landing operation. For cases used for operation, the unit shall be capable of being operated outdoors in a moderate rain with covers removed when placed on the ground or when mounted on a rotating pedestal. (See 3.8.2, 3.8.4.)

3.8.2 Case, Receiver-Transmitter CY-944( )/TPX.- This case, constructed in accordance with model, shall house the Receiver-Transmitter Radio RT-211( )/TPX for operation and transit and furnish mounting facilities for Coder-Control, Interrogator Set KY-97( )/TPX.

3.8.3 Case, Standardized Components, Electrical CY-951( )/TPX.- This case, constructed in accordance with model, shall contain either Coder-Control, Interrogator Set KY-97( )/TPX and associated spare parts and cables as called for in the bid request and contract or the associated spare parts of Coder-Control, Interrogator Set KY-99( )/UPX as called for in the bid request and contract. This case is not designed to contain the Coder-Control, Interrogator Set KY-99( )/UPX.

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3.8.4 Case, Receiver-Transmitter CY-1324( )/UPX-6.- This case, constructed in accordance with model, shall house the Radar Recognition Set AN/UPX-6 for operation and transit.

3.9 Receiver-Transmitter, Radio RT-211( )/TPX.- The receiver-transmitter shall conform to the following:

3.9.1 Adjustment and repair.- The receiver-transmitter shall be capable of continuous use with Radar Test Set AN/UPM-4, or with Radar Test Set AN/UPM-6B. The monitoring provisions in the equipment shall provide for the use of Radar Test Set, AN/UPM-4 as the measuring and indicating instrument. Provisions shall be made for the use of Radar Test Set AN/UPM-6B, as a "go-no-go" indicator.

3.9.2 Combination dummy load attenuator.- The plug-in type dummy load attenuator in accordance with model shall, when used as a dummy load, have an impedance of 52 ohms and shall present a standing wave ratio no greater than 1 db over the band from 1005 to 1055 mc, and shall be capable of dissipating an average power of 3 watts. When used as an attenuator it shall provide 12 db ± 1 db attenuation.

3.9.3 Direction r-f probe.- For external monitoring an r-f probe of the directional type, coupled to the antenna circuit of the receiver-transmitter, shall be provided for test purposes. This probe shall provide coupling to the forward power and shall be provided with a type BNC connector for connection to external equipment. When feeding into a 52-ohm line with a standing wave ratio of not over 2 db, the probe shall present an attenuation of 20 ± 0.5 db at the center of the band from 1005 to 1055 mc and 20 ± 1.5 db throughout this band to the forward-going power. The attenuation of the probe shall be at least 18 db greater to reflected power than to power in the forward direction regardless of the phase between the two directions of power. When terminating a 52-ohm line, the probe shall not produce a standing wave ratio of greater than 2 db. A label on the front panel adjacent to the BNC connector shall be provided indicating "Mid-band attenuation 20 db."

3.9.4 Construction.- The construction shall be in accordance with model unless otherwise specified herein.

3.9.5 Performance.-

3.9.5.1 Receiving circuit.- The receiving circuit shall meet the following requirements:

3.9.5.1.1 Local oscillator frequency.- The receiver's local oscillator shall operate at a frequency differing from that of the received signal by an amount equivalent to the mid-frequency of the i-f band as specified in 3.9.5.1.4.

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3.9.5.1.2 Sensitivity. - The sensitivity of the receiver is defined as the rms voltage during the pulse of 1 microsecond duration, required to give signal plus rms noise to rms noise ratio of 11 to 1, at the frequency of maximum response in any pass band between 1065 and 1115 mc, expressed in decible below one volt using a signal generator calibrated in terms of open circuit voltage and having a nominal impedance of 52 ohms. This sensitivity shall be 76 db below one volt or better and hereafter will be designated as "Normal Sensitivity." The overall noise figure of the receiver shall not exceed 16 db. For convenience the receiver sensitivity may be measured by means of an approved type noise figure measuring diode type of equipment and an indicating device connected to the output of the last i-f amplifier stage. The indicating device must respond to the rms value of noise and signal.

3.9.5.1.3 Overall band-width. - The band-width of the receiver 6 db down form the maximum response with the gain set as in 3.9.5.1.12 and with one half volt output shall normally be between 8 and 11 megacycles, and not more than 24 megacycles at 40 db down from maximum response as measured with a c-w signal generator. The top of the overall selectivity curve shall be flat to within $\pm$1.0 db and shall not vary appreciably with different gain control settings. It shall be possible by a suitable switching means located on the main chassis of the receiver-transmitter to narrow the bandwidth in such a way as to afford 70 db adjacent channel (2.5 mc) rejection. This may be accomplished by means of the so called McKinley discriminator.

3.9.5.1.4 Intermediate frequency. - The mid-frequency between points 6 db down from the maximum response of the intermediate amplifier shall be 60.0 $\pm$ 0.5 megacycles.

3.9.5.1.5 I-F gain. - When the receiver-transmitter is connected to the Coder-Control (KY-97( )/TPX) the gain control potentiometer in the control box shall be connected in the receiver circuit at a point such that the amplification between i-f input and the point of control does not exceed 40 db. The sum of the attenuation produced by the gain control and the gain-time control (3.9.5.1.6) need not exceed 80 db.

3.9.5.1.6 Gain-time control and gate. - An electronic switch shall be provided which, when not being triggered, shall keep the receiver gain at least 75 db below the "normal" gain (that gain required to produce the output of 3.9.5.1.12). When the electronic switch is triggered, the receiver gain shall rise from the 75 db down value to an adjustable value between 20 and 35 db down. The gain shall then continue to return to "normal" in accordance with one of the three "Conditions" of 3.9.5.1.6.1. After the gain has been returned to "normal" for a preset interval, it shall again drop to the untriggered value of 75 db below "normal" gain.

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3.9.5.1.6.1 Rise to normal gain.- After the electronic switch has been triggered by a pulse coincident with the first transmitted pulse, the receiver gain shall rise to a value between 20 and 35 db below "normal." The value between these figures shall be adjustable by one or more controls inside the receiver-transmitter cabinet. From the value so adjusted (typical "settings" of which are given in 3.9.5.1.6.1.1 and 3.9.5.1.6.1.2) the gain shall continue toward "normal" in accordance with one of the three "Conditions" given below. Conditions "1", and "2", shall be selectable remotely by operations of low voltage control circuits with controls on the front panel of the coder-control. The various "Conditions" shall be obtainable while the transmitter is delivering normal power to its antenna terminal. The mode switch shall be set to Interrogation Mode No. 1 when test of GTC characteristics are being conducted.

3.9.5.1.6.1.1 Condition one.- The gain shall rise in accordance with the following data:

Risetime (after 1st transmitter pulse)	Setting I	Setting 2	Setting 3
10 Microseconds	35 db	25 db	20 db
100 Microseconds	20 db		
200 Microseconds	14 db		
400 Microseconds	8 db	0 db	0
600 Microseconds	0-4 db		
1000 Microseconds	0 db	0 db	0

3.9.5.1.6.1.2 Condition two.- The gain shall rise in accordance with the following data:

Risetime (after 1st transmitter pulse)	Setting I	Setting 2	Setting 3
10 Microseconds	35 db	26 db	20 db
75 Microseconds			0
100 Microseconds	12 db	0	
175 Microseconds	0	0	0

The tolerances for the attenuation values in the above tables are + 10 percent and -20 percent of the db values of 3 db, whichever is greater.

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3.9.5.1.6.2 Gate length and receiver gain.- After the gain reaches its normal value under Conditions 1 or 2, the gain shall remain within 1 db of its normal value for a time interval of continuously adjustable length such that the total duration from the start of the second pulse shall be adjustable over the range from 300 to 2500 microseconds, however, the interval need not be shorter than the gain rise time for any given adjustment of the rise time. It shall be possible to preset the length of this gate by means of installation adjustments inside the equipment. At the end of the gate time the gain of the receiver shall fall to at least 75 db below its normal value as rapidly as practicable but within 100 microseconds. Prior to acceptance, the contractor shall set the gate durations to 720 microseconds.

3.9.5.1.7 Gate and GTC off.- A shorting link shall be provided which when in the "off" position will make the sensitivity of the receiver independent of the gate and the gain time control circuit.

3.9.5.1.8 Anti-jamming provisions.- The receiver shall be capable of operating through jamming as follows:

3.9.5.1.8.1 A pulse signal having a voltage of 84 db below 1 volt or weaker shall be perceptible through c-w jamming of 66 db below 1 volt or stronger.

3.9.5.1.8.2 With the gain control set to give 0.03 volt rms noise in the absence of c-w, a c-w signal voltage of 72 db below 1 volt or stronger having 30 percent modulation at 1000 cycles shall produce not more than 1.0 volt peak to peak at the receiver output, corresponding to the 1000-cycle modulation envelope and its harmonics.

3.9.5.1.9 Fidelity and recovery characteristics of the receiver.-

3.9.5.1.9.1 A signal as described in 3.9.5.1.9.1.1 impressed at the input of the receiver with the gain control setting to give noise output voltage of 0.06 rms volt shall produce a signal as described in 3.9.5.1.9.1.2 across 75 ohms at the output of the receiver.

3.9.5.1.9.1.1 Input signal.-

(a) Duration, microsecond or less.

(b) Amplitude, sufficient to produce one-half saturation at the output of the receiver.

(c) Time of rise, not greater than 0.15 microsecond.

(d) Time of decay, not greater than 0.15 microsecond.

(e) Slope of pulse, not greater than 2 percent.

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3.9.5.1.9.1.2 Output signal.-

(a) Duration, 0.9 to 1.1 microseconds.

(b) Amplitude, not greater than one-half saturation level.

(c) Time of rise, not greater than 0.4 microsecond.

(d) Time of decay, from 60 to 10 percent maximum amplitude shall be not greater than 0.4 microsecond.

(e) Slope of pulse, shall be not greater than 40 percent.

3.9.5.1.9.2 A signal as described in 3.9.5.1.9.2.2 impressed at the input of the receiver in such a manner that its leading edge occurs 4 microseconds after the leading edge of a signal as described in 3.9.5.1.9.2.1 shall produce a signal as described in 3.9.5.1.9.2.3 across 75 ohms at the output of the receiver when the gain is set for 0.03 volt rms noise.

3.9.5.1.9.2.1 Input signal No. 1.-

(a) Duration 2.5 to 2.7 microseconds.

(b) Amplitude, 60 db or less below 1 volt.

(c) Time of rise, not greater than 0.15 microsecond.

(d) Time of decay, not greater than 0.15 microsecond.

(e) Slope, not greater than 2 percent.

3.9.5.1.9.2.2 Input signal No. 2.-

(a) Duration, 1 microsecond or less.

(b) Amplitude, 74 db below 1 volt.

(c) Time of rise, not greater than 0.15 microsecond.

(d) Time of decay, not greater than 0.15 microsecond.

(e) Slope, not greater than 2 percent.

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3.9.5.1.9.2.3 Output signal produced by input signal No. 2.-

(a) Duration 1.0 to 1.1 microsecond

(b) Amplitude, at least 0.25 volt

(c) Time of rise from 10 to 70 percent maximum amplitude shall not be greater than 0.4 microsecond.

(d) Time of decay, not greater than 0.4 microsecond.

(e) Slope, not greater than 30 percent.

3.9.5.1.10 Receiver delay.- The delay between the envelope of an input r-f pulse of approximate y 70 db below 1 volt with the gain set so as to give 0.03 volt of rms noise, and the video output pulse, which is derived at the video output of the receiver when the r-f pulse is impressed at the antenna terminal, shall be not greater than 0.5 microsecond.

3.9.5.1.11 Intermediate-frequency pickup.- The receiver circuit shall be substantially shielded from energy pickup on any frequency in the i-f band. Sensitivity to energy within this band impressed at the antenna terminal shall be at least 60 db below the sensitivity at the tuned frequency of the receiver.

3.9.5.1.12 Video output level.- With the gain set to produce 0.03 volts rms noise, a signal level of not less than 4 volts (12.04 db above 1 volt) positive shall be obtainable. The output shall limit at a voltage not greater than 7 volts across an output impedance of 75 ohms. The dynamic range of the output shall be at least 20 db. It shall be possible to remove the circuit which produces limiting by means of an internal installation adjustement.

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3.9.5.1.13 Frequency stability and band width decrease.- The receiver local oscillator frequency shall not exceed ± 200 kilocycles of assigned frequency and the overall bandwidth of the receiver shall not decrease by more than 1.0 megacycle under all combinations of the following conditions which can reasonably be expected to occur during a 24-hour interval at a field station, at a shore base, in a plane, or aboard a ship where the receiver-transmitter can reasonably be expected to be installed:

(a) Ambient humidity from 20 percent to the dew point at 122° F.

(b) Line voltage variation of ± 10 percent from the nominal value.

(c) Variation in barometric pressure ± 1 inch from a nominal value within the range from 25 to 31 inches of mercury.

(d) Variation in room ambient temperature to within the range from -40° to 131° F.

(e) Minimum to maximum settings of manual IF gain control.

Before conducting tests to determine if the provisions of this paragraph are being met, the equipment shall be permitted to warm up for an interval of ten minutes if initially in un-energized condition.

3.9.5.1.14 Receiver tuning.- The receiver shall operate at any selected frequency (determined by the crystal used) and the setting of the receiver tuning control in the frequency band from 1065 to 1115 megacycles.

3.9.5.1.14.1 Receiver crystals.- One receiver crystal shall be provided with each equipment for controlling and setting the operating frequency of the receiver to 1090 mc ± 200 kc.

3.9.5.1.15 Sensitivity drift.- The sensitivity of the receiver at the center frequency of the pass band shall not be reduced by more than 3 db and the gain of the receiver shall not vary by more than plus or minus 6 db from the specified magnitudes by any combination of the conditions specified in 3.9.5.1.13.

3.9.5.1.16 Local oscillator shielding.- The signal delivered by the local oscillator into a 52 ohm resistive load terminating the antenna terminal of the receiver-transmitter equipment shall be not stronger than 40 db below one volt.

3.9.5.1.17 Image ratio.- The image ratio of the receiver shall be not less than 70 db.

3.9.5.1.18 S-band and X-band rejection.- The sensitivity of the receiver to pulse signals whose carrier frequency is in the S- and X- bands shall be at least 40 db below the sensitivity to signals in the band 1005 to 1115 mc.

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3.9.5.1.19 Line voltage fluctuation. - Transient signals produced by reason of line, voltage fluctuation of ± 15 percent of the impressed voltage for durations up to 5 seconds, shall not deleteriously affect the equipment nor result in a base line shift greater than 10 percent of the saturation value of the video output of the equipment as measured across a 75-ohm load resistor.

3.9.5.1.20 Overall receiver gain. - The overall receiver gain throughout the tuning range specified in 3.9.5.1.14 shall be such that a 3.0 microvolt pulse signal (with the pulse shape characteristics of input signal No. 2 described in 3.9.5.1.9.2.2) applied to the antenna connector on the front panel of the equipment shall produce a 2-volt pulse at the video output connector on the front panel.

3.9.5.2 Performance of receiver-transmitter transmitting circuit. - The transmitting circuit shall meet the following requirements:

3.9.5.2.1 Power output. - The transmitter oscillator shall be capable of delivering no less than1500 watts r-f power as measured on a suitable meter such as the one on the AN/UPM-4 Test Set after calibration. Such power shall be delivered at any desired frequency between 1005 and 1055 mc to a 52-ohm resistive termination at the antenna connector on the equipment at nominal line voltage and at any recurrance rate from 200 to 400 pulses per second, and with reduced power up to 1000 pulses per second. The power output of the transmitter shall not be reduced by more than 2 db due to the cause given in 3.9.5.2.1.1. Moreover, the power output of the transmitter shall not be reduced by more than an additional 2 db over the condition defined in 3.9.5.1.13. Provision shall be made for reducing power output by approximately 12 db.

3.9.5.2.1.1 The transmitter shall deliver the specified power, with all values of load impedance which present a standing wave ratio up to and including 3.5 db at the antenna terminal of the receiver-transmitter, with the incident and reflecting waves coincident over at least 98 percent of their duration.

3.9.5.2.2 Tuning. - The transmitter shall operate on any selected frequency (determined by the crystal used) in the frequency band from 1005 to 1055 megacycles.

3.9.5.2.3 Frequency stability. - The transmitter frequency stability, with a mismatch from all causes such as crystal drift, switching, etc., at the antenna causing a standing wave ratio as great as 3.5 db, shall be not greater than ± 200 kilocycles. (see 3.10.2.1.2, 3.9.5.3.2.)

3.9.5.2.4 Transmitter crystals. - One crystal shall be provided with each equipment for controlling and setting the frequency of the transmitter to 1030 mc ± 200 kc.

3.9.5.2.5 Transmitted pulse duration. - The duration of any r-f pulse generated by the transmitter shall be between 0.7 and 1.2 microsecond under any combination of the conditions specified in 3.9.5 1 13.

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3.9.5.2.6 Duty cycle capability.- The transmitter shall operate without overheating or loss of power below 1500 watts or any r-f duty cycle up to 0.15 percent. The transmitting equipment shall be capable of delivering a train of 16 one-microsecond pulses in a period of 50 microseconds with alternate pulses spaced as close as 3 microseconds to the preceeding pulse. In a pulse train of 16 pulses, no pulse amplitude shall vary more than 10 percent from the first pulse in the train. The equipment shall be capable of transmitting trains of 16 pulses at a repetition rate of 400 pulse trains per second for 5 successive repetition periods after which the transmitter need only transmit a pair of pulses for a sufficient period to make the average duty cycle 0.15 percent.

3.9.5.2.7 R-F pulse shape.- The rise time of each r-f transmitted pulse developed shall be not greater than 0.20 microsecond nor shall the time decay be greater than 0.4 microsecond measured with a suitable low-distortion detector similar to the one incorporated in the Radar Test Set AN/UPM-4 equipment. The minimum voltage amplitudes of the tops of the two successive pulses of a pair shall be within 10 percent of each other. The form of the pulse tops shall be in accordance with 3.7.4. The instantaneous voltage amplitudes of the pulses shall not fluctuate more than 15 percent due to ripple voltages generated within the equipment.

3.9.5.2.8 Transmitter delay.- The delay between the leading edge of a pulse delivered to the modulator when a trigger pulse of 25 volts is applied, and the leading edge of the r-f envelope of the resulting transmitter pulse shall be not greater than 0.8 microsecond. The jitter in any pulse of a train of 16 resulting transmitter pulses shall not exceed 0.06 microsecond.

3.9.5.2.9 Transmitter tube change.- On replacing the tubes in the transmitter circuits in an equipment which was in normal operation prior to the replacement, the adjustments provided shall be sufficient to bring the circuit back into peak operation.

3.9.5.2.10 Transmitter monitor.- The transmitter monitor shall comprise a lower monitoring device built into the equipment. The sensitivity adjustment shall compensate for changes in power level. This device shall cause the transmitter monitor pilot light to go off if there is a drop in transmitter power output of 3 to 6 db and more.

3.9.5.3 Overall performance of receiver-transmitter.- The receiver-transmitter shall meet the following requirements:

3.9.5.3.1 Trigger out pulse.- The equipment shall provide a trigger out pulse to a 75-oh load of an amplitude of between 20 and 50 volts peak positive, a duration of from 1.0 to 3.0 microseconds and a time of rise of not more than 0.25 microsecond, 0 to 1.0 microsecond ahead of the first r-f pulse of each pulse train. The trailing edge shall have no more of negative overshoot than 20 percent of the positive amplitude when working into the 75-ohm. This pulse shall also be produced when the equipment is operating in the free run condition.

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3.9.5.3.2 Frequency controls. - The transmitter and receiver each shall have a single tuning control. All transmitter and receiver circuits which must be retuned when frequency determining crystals are changed shall be continuously adjustable over the operating frequency range by the single frequency controls. The single frequency controls shall be calibrated directly in operating frequency. The accuracy of these calibrations shall be such that specified transmitter and receiver performance will be obtained by setting the control exactly to the operating frequency reading Means shall be provided for plugging in the transmitter and receiver frequency control quartz crystals. After readjustment of the controls, the transmitter and the receiver local oscillator shall be on the assigned frequency with a tolerance of ± 200 kc. The transmitter as well as the receiver local oscillator shall be direct crystal controlled. Automatic frequency control of either of the transmitter or receiver is not acceptable. The tuning of the receiver over the assigned frequency band shall not change the frequency to which the transmitter is tuned nor shall the tuning of the transmitter over the assigned frequency band change the frequency to which the receiver is tuned.

3.9.5.3.2.1 Quartz crystals. - The transmitter and receiver frequency control quartz crystals shall be Type CR-23/U with Type HC-6/U crystal holder, per Specification MIL-C-3098.

3.9.5.3.3 Transmitter-receiver frequency separation. - The receiver shall operate without deterioration in the performance specified with a frequency difference of 40 megacycles or greater between the transmitter and receiver. The receiver mixer crystal shall not be damaged by transmitter and receiver operation with 40 megacycles of each other with the antenna connected or disconnected.

3.9.5.3.4 Common antenna terminal. - The common antenna terminal provided per model for both transmitter and receiver operation shall be terminated with a 52-ohm resistive load.

3.9.5.3.5 Character of interrogation. - The transmitter shall be capable of delivering coded radio-frequency pulse trains as follows at its antenna terminal at any frequency in the band from 1005 to 1055 mc.

3.9.5.3.5.1 Interrogation Mode 1. - This mode shall consist of a pair of pulses spaced 3.0 ± 0.2 microseconds between leading edges. (See 3.10.2.2 6.1)

3.9.5.3.5.2 Interrogation Mode 2. - This mode shall consist of a pair of pulses spaced 5.0 ± 0.2 microseconds between edges. (See 3 10.2.2 6.2)

3.9.5.3.5 3 Interrogation Mode 3. - This mode shall consist of a pair of pulses spaced 8.0 ± 0.2 microseconds between leading edges. (See 3.10.2.2.6.3).

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3.9.5.3.6 Character of reply.- The receiver-transmitter shall be capable of receiving reply signals as below at its antenna terminal at any frequency from 1065 to 1115 megacycles and delivering their envelopes to its receiver output terminals:

(a) Pulses between 0.9 to 1.3 microseconds in width. Their rise time and decay shall be as specified in 3.10.2.2.6.1.

(b) Trains of two one-microsecond pulses as specified in (a), the spacing between the leading edges of which shall be 16 ± 2.5 microseconds.

(c) Trains of four one-microsecond pulses each as described in (a), the spacing between the leading edges of successive pulses being 16 ± 2.5 microseconds.

(d) Trains of 16 or less successive pulses as described in (a), spaced 3 microseconds between pulses.

3.9.5.3.7 Overload relay protection.- Following an overload, the overload protection circuit shall remove the plate voltage from those circuits which may be damaged from the overload. Following the overload, the circuit shall automatically reset to normal operation. If the overload persists long enough for the resetting action to take place for 1 to 2 seconds, the resetting shall cease. After resetting action has ceased the voltage supply to the remote indication lamps shall be removed and further resetting of the overload protection circuit shall then be accomplished by opening the transmitter on-off switch.

3.9.5.3.8 Power consumption.- The total power consumed, as supplied and measured at the power input terminal of the equipment shall not exceed 250 watts or 300 volt-amperes. The equipment shall meet this power requirements when in normal operation "challenging" at a rate of 400 pulse pairs per second, and with all electro-mechanical components performing their appropriate functions.

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3.9.5.3.8.1 Power supply. - The receiver-transmitter shall operate from single phase power supplies having a voltage regulation of ± 10 percent and normal line voltages and frequencies as follows:

Voltage-root-mean-square volts	Frequency in cycles per second
117 ± 10 percent	400 ± 10 percent
105 ± 10 percent	400 ± 10 percent
125 ± 10 percent	400 ± 10 percent

The alternating supply voltage may have a form factor within the range 1.25 to 1.02 and may have an amplitude factor within the range 1.68 to 1.16. Form factor is defined as the ratio of the root-mean-square voltage to the average voltage and has a value of 1.11 for a true sine wave. Amplitude factor is defined as the ratio of the peak instantaneous voltage to the root-mean-square voltage and has a value of 1.414 for a true sine wave. Line supply surges of 350 volts persisting for one cycle shall cause no permanent damage to the equipment.

3.10 Coder-Control, Interrogator Set KY-97( )/TPX. - Coder-Control KY-97()/TPX shall conform to the following:

3.10.1 Construction. - The construction shall be in accordance with model unless otherwise specified herein.

3.10.2 Performance. -

3.10.2.1 General. -

3.10.2.1.1 Manual i-f gain control. - The gain control shall be 1000 ohm potentiometer. The gain of the interconnected receiver expressed as a function of the angle of rotation of the gain control knob shall increase approximately logarithmically over a range of at least 40 db in the presence of gain-time control, or 55 db in the absence of GTC, as the knob is rotated clockwise.

3.10.2.1.2 Challenge switch. - In not more than one second after this switch is manipulated from the "off" to the "on" position the interconnected interrogator shall be in full operation with the frequency stabilized to within ± 200 kc of its proper operating frequency. (See 3.5.9.2.3) Transmitter monitor pilot light on receiver-transmitter and coder-control shall go "on" except under conditions stated in 3.9.5.2.10.

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3.10.2.2 Detail.-

3.10.2.2.1 Triggering.- The control circuits shall operate with a trigger input having the following characteristics:

(a) Polarity: positive only.

(b) Pulse width: 0.3 to 15 microseconds at half amplitude.

(c) Amplitude: 5 to 50 volts measured across 75-ohm terminating resistor at the input to the coder-control.

(d) Pulse repetition frequency: up to 4100 pps.

(e) Time of rise: from 0 to 70 percent of full amplitude within 20 percent of pulse length, but in any case between 0.1 and 0.5 microsecond.

3.10.2.2.1.1 Self-trigger.- When the self trigger switch within the equipment is operated to "on" position, the coder-control unit shall operate independently at a repetition rate of 275 ± 125 pps. When self triggered the coder-control shall deliver to the transmitter such pulses as are required for otherwise normal operation. The operation of receiver-transmitter shall not be affected by an open circuit across the trigger output connector.

3.10.2.2.2 Count down.- A count down circuit shall be provided which shall be capable of maintaining the transmitter's repetition rate below a preset value when trigger pulses with a repetition rate as high as 4100 pps are supplied. The preset value shall be continuously adjustable between 200 and 1000 pps by means of a chassis control.

3.10.2.2.3 Trigger delay.- The total delay between a trigger pulse and the second pulse of the r-f pulse train associated therewith, shall be identical on all modes of interrogation and shall be capable of adjustment to a value between 12.5 ± 0.5 microseconds and 35 ± 2.0 microseconds as preset by a continuously adjustable internal control. An internal chassis switch per model shall be capable of removing the delay circuit.

3.10.2.2.4 Coder.- When triggered, the coder shall deliver such pulses to the transmitter modulator as are required to produce trains of r-f pulse pairs as specified in 3.10.2.2.6. The pulses delivered from the coder shall be delayed by a different time interval for each mode of operation.

3.10.2.2.5 Control circuit output.- When triggered by pulses as specified in 3.10.2.2.1 the coder circuit shall deliver to the transmitter such pulses as are required to produce trains of r-f pulses as specified in 3.10.2.2.6.

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3.10.2.2.6 Modes of interrogation.- When a single trigger pulse is applied to the control circuits, any one of the three following modes of transmitted trains delayed as stated in 3.10.2.2.3 of r-f pulses shall be available by use of the selector switch.

3.10.2.2.6.1 Interrogation Mode No. 1.-

(a) Duration of envelope: 0.7 to 1.2 microseconds.

(b) Duration between envelopes: 3.0 ± 0.2 microseconds.

(c) Time of rise of envelope: not greater than 0.2 microsecond.

(d) Time of decay of envelope: not greater than 0.4 microsecond.

3.10.2.2.6.2 Interrogation Mode No. 2.-

(a) Duration of envelope: 0.7 to 1.2 microseconds.

(b) Duration between envelopes: 5.0 ± 0.2 microseconds.

(c) Time of rise of envelope: no greater than 0.2 microsecond.

(d) Time of decay of envelope not greater than 0.4 microsecond.

3.10.2.2.6.3 Interrogation Mode No. 3.-

(a) Duration of envelope: 0.7 to 1.2 microseconds.

(b) Duration between envelopes: 8.0 ± 0.2 microseconds.

(c) Time of rise of envelope: not greater than 0.2 microsecond.

(d) Time of decay of envelope: not greater than 0.4 microsecond.

3.10.2.2.7 Interrogation mode change.- The coder-control shall be so arranged that, upon applying one triggering signal in accordance with 3.10.2.2.1, it will be possible to obtain any one of the transmitted trains specified in 3.10.2.2.6 when so selected at the election of the operator.

3.10.2.2.8 Chop rate.- The chop rate shall be adjustable from 3 to 23 cps.

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3.11 Coder-Control, Interrogator Set KY-99( )/UPX.-

3.11.1 Construction.- Construction shall be in accordance with model, unless otherwise specified herein.

3.11.1.1 Waterproof cabinet.- The coder-control shall be enclosed in a cabinet which will render it waterproof and capable of being floated ashore during landing operations.

3.11.2 Performance, requirements for the Coder-Control, Interrogator Set KY-99( )/UPX.- The control circuits shall meet the following requirements:

3.11.2.1 Triggering.- The control circuits shall operate with a trigger input having the following characteristics:

a. Polarity: positive.

b. Amplitude: 10-30 volts measured across a 510 ohm terminating resistor.

c. Pulse repetition frequency: Swept at random rate between 3600 to 4000 pps.

d. Pulse width: 0.5 to 3.0 sec. at 50 percent amplitude.

e. Rise time: between 0.1 and 0.4 microsecond.

3.11.2.2 Self trigger.- When the self-trigger switch within the coder-control is thrown to the "on" position, the coder-control unit shall operate independently at a repetition rate of 275 ± 125 pps. When self-triggered, the coder-control shall deliver to the transmitter such pulses as are required for otherwise normal operation. The operation of this unit and its associated equipment shall not be adversely affected by an open circuit across the trigger output connector. (See 3.10.2.2.1.)

3.11.2.3 Count down.- The count down circuit shall be capable of maintaining the transmitter's repetition rate of approximately 420 pps when trigger pulses with repetition rate of 3800 (count down of 9) are supplied.

3.11.2.4 Coder.- When triggered, the coder shall deliver such pulses to the transmitter modulator as are required to produce trains of r-f pulse pairs as specified in 3.10.2.2.6. The delay between the input trigger pulse and second output pulse of the coder shall be 10 ± 0.2 microseconds on all modes of interrogation. Output pulses from the coder shall have an amplitude of at least 30 volts across 75 ohms.

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3.11.2.5 Control circuit output. - When triggered by pulses as specified in 3.11.2.1 the coder circuit shall deliver to the transmitter such pulses as are required to produce trains of r-f pulses as specified in 3.10.2.2.6.

3.11.2.6 Interrogation mode change. - The coder shall be so arranged that, upon applying one triggering signal in accordance with 3.11.2.1, it will be possible to obtain any one of the transmitted trains specified in 3.10.2.2.6 when so selected at the election of the operator. The parameters which determine the pulse spacings, if adjustable, shall be suitably locked to prevent changes under operating service conditions.

3.11.2.7 Challenge switch. - Challenging of the color-control shall be initiated by the closing of an external switch to ground. Immediately after this switch is thrown from the "off" to the "on" position, the inter-connected interrogator shall be in full operation.

3.11.2.8 Input voltage. - The coder-control shall operate from the following voltages:

a. -300 volts regulated

b. +200 volts regulated

c. 117 volts rms ± 10% at a frequency of 60 cps ± 10%

3.12 Bounce conditioning. - After final assembly, each component on contract, with shock mounts removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this conditioning for 1 minute. After conditioning the equipment shall be capable of meeting the inspection of section 4.

3.13 Seal test, provision for. - A pluggable hole shall be provided in the transit case, or in the equipment inclosure when no transit case is employed, for use in performing the seal test (See 4.7.2)

3.14 Technical literature, tools, and running spare parts. - Technical literature, tools, and running parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

3.15 Workmanship. - Components, parts, sub-assemblies, and complete equipment shall be manufactured and assembled in a thoroughly workmanlike manner.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection, general.- Government inspection shall consist of group A, group B, and group C inspection as specified in 4.1.1 through 4.1.3.2. (See 6.3.1.) Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection, or both.

4.1.1 Group A inspection.- Group A inspection shall be in accordance with table IV and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector, except that the air-seal test (see 4.2) shall be last.

4.1.2 Group B inspection.- Group B inspection shall be in accordance with table IV.

4.1.2.1 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.2.2 Sampling procedure.- Sampling plans for group B inspection shall be as specified in the contract and shall be in accordance with the applicable drawings specified in table III. Samples shall be selected in an unbiased manner by the Government inspector. Unless otherwise specified, inspection shall be started using level II of the specified drawing. If the number of defective sample units from a lot exceeds the acceptance number specified on the drawing, the lot shall be rejected. Disposition of rejected product (sample units and lots) shall be in accordance with Standard MIL-STD-105 and 4.1.5.

4.1.2.3 Procedure in case of failure.- When a lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and proposed corrective action. If the contractor and Government inspector cannot agree on the corrective action to be taken, the matter shall be referred to the contracting officer for resolution.

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Table III. - Drawings for Group B inspection plans

Group B plan	Drawing
B-1	460-11
B-2	460-12
B-3	460-13
B-4	460-14

4.1.3 Group C inspection. - This shall consist of the group C inspection specified in table IV.

4.1.3.1 Selection of samples. - Two equipments for each group C inspection shall be selected each month in an unbiased manner by the Government inspector, unless otherwise specified herein (see 4.12). Additional samples shall be selected when deemed necessary by the contracting officer.

4.1.3.1 Noncompliance. - If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and proposed corrective action to be taken on the process and affected portion of product. If the Government inspector does not consider that the corrective action will enable the equipment to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.1.4 Reinspection of conforming group B and group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract provided that all visible damage incurred during group B or group C inspection is repaired and the units are then subjected to and pass group A inspection

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4.2.5 Disposition of nonconforming product.- When defective sample units or rejected lots are resubmitted or affected portion of product is submitted for acceptance, such items shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government.

Table IV.- Inspection procedures

Inspection	Rqt par.	Test par.	Group A AQL	Group B Plans	Group C
Vibration	3.5.3.1	4.2			X
Shock; bench handling	3.5.3.2	4.4			X
Shock; drop	3.5.3.2	4.5.1		X	
Shock; drop	3.5.3.2	4.5.2		X	
Air seal	3.13	4.7.2	1%		
Temperature	3.5	4.8			X
Variation in supply line voltage	3.9.5.1.20 3.9.5.3.7.1	4.11		X	
Change of tubes	3.9.5.2.7 3.9.5.1.21	4.12			X
Microphonic tubes	3.9.5.1.13	4.13	1%		
Twenty-four hour performance test	3.9.5.1.13	4.14	1%		
Corona test		4.15		X	
Receiver performance after 24 hour test run		4.16			
Sensitivity	3.9.5.1.2	4.16.1	Note 1		
Image ratio	3.9.5.1.3	4.16.2			X
Bandwidth	3.9.5.1.3	4.16.3		X	
Output level	3.9.5.1.12	4.16.4	Note 1		
Video output	3.9.5.1.12	4.16.5	1%		
I-F- gain control	3.9.5.1.5	4.16.6	1%		
Gain-time control and gate	3.9.5.1.6	4.16.7	Note 1		
Anti-jamming	3.9.5.1.8	4.16.8			X
Fidelity and recovery	3.9.5.1.9	4.16.9			X
Receiver delay	3.9.5.1.10	4.16.10			X
I-F pickup	3.9.5.1.11	4.16.11			X

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Table IV.- Inspection procedures

Inspection	Rqt par.	Test par.	Group A AQL	Group B Plans	Group C
Transmitter and Coder Control, Interrogator Set KY-97( )/TPX performance After 24 hour test run		4.17			
Power output	3.9.5.2.1	4.17.1	Note 1		
R-F pulse shape	3.9.5.2.7	4.17.2	Note 1		
Transmitter delay	3.9.5.2.8	4.17.3		X	
Trigger out pulse	3.9.5.3.1	4.17.4	1%		
Operation		4.17.5		X	
R-F- pulse spacings	3.10.2.2.6	4.17.6	Note 1		
Chop rate	3.10.2.2.8	4.17.7	1%		
Performance of Coder Control KY-99( )/UPX	3.11	4.18	1%		
Twenty-four hour performance	3.11	4.18	1%		
Visual and Mech. Major		4.19	1%		
Visual and Mech. Minor		4.19	4%		

NOTE 1: All items marked Note 1 shall be considered a group. The AQL for the group shall be 1%.

4.2 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.5.3, as follows:

a. The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration. (See 4.2(d).)

b. The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be constant at 1/64 inch.

c. Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

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d. If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because the unit weighs more than 250 pounds or because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.3 Bounce test.- The applicable bounce test as specified for components shall be applied as follows:

4.3.1 Preproduction test.- The Receiver-Transmitter and Coder-Control, Interrogator Set KY-97( )/TPX, in their respective transit cases, shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester shafts in phase, shall be operated at a speed of 285 r.p.m. ±1 percent, for a total of 3 hours. During each 1/2-hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.3.2 Preproduction test.- The Coder-Control, Interrogator Set KY-99( )/UPX, as packaged for shipment, shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 r.p.m. ±1 percent, for a total of 3 hours. During each 1/2-hour period of the test, the packaged equipment shall rest on a different face.

4.3.3 Preproduction test.- The sample of Coder-Control, Interrogator Set KY-99( )/UPX, specified in 3.3.1, shall be subjected to this test only. The Coder-Control, Interrogator Set KY-99( )/UPX, on its shock mounts (if any), shall be secured to the Vehicular Adapter Plate similating actual usage using mounting provided, and placed on the table of the Package Tester, both as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The plate shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 r.p.m. ±1 percent, for a total of 3 hours. The adapter plate shall be rotated through 90 degrees, in the same direction, at the end of each 3/4-hour period.

4.4 Shock test; bench-handling.- The receiver-transmitter, and two coder-controls shall be subjected to this test. The chassis-and-front-panel assembly shall be removed from its inclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liabel to occur during servicing)

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a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as a pivot, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as pivots, for a total of four drops.

b. Repeat "a", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly could be placed practicably during servicing.

3.4.5 Shock tests. - The applicable shock test shall be applied as specified.

4.5.1 Shock test; drop. - The receiver-transmitter and Coder-Control, Interrogator Set KY-97( )/TPX, in their respective transit cases, shall be dropped a distance of 4 feet on each corner, edge, and face (total of 26 drops.) Drops shall be made from a quick-release hook; an Acme Drop Tester, or equal; a Conbur Tester (10-Degree inclined plane), or equal; or a basket-type pendulum tester. The floor or barrier receiving the impact shall be 2-inch fir, backed by concrete or a rigid steel frame.

4.5.2 Shock test; drop. - The Coder-Control, Interrogator Set KY-99( )/UPX, as packaged for shipment, shall be dropped a distance of 30 inches on each corner, edge, and face (total of 26 drops.) Drops shall be made from a quick-release hook; an Acme Drop Tester, or equal; a Conbur Tester (10-degree inclined plane), or equal, or a basket-type pendulum tester. The floor or barrier receiving the impact shall be 2-inch fir backed by concrete or a rigid steel frame.

4.6 Shock test; ballistic. - The receiver-transmitter and two Coder-Control in operating cases, (if any), shall be subjected to this test. The test shall be conducted on the "Shock Testing Machine for Light Weight-Equipment" as shown in Specification MIL-S-901. The equipment, including shock mounts (if any), shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 6 blows: one each 1-foot blow, and 3-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible.

4.7 Immersion tests. - These tests shall be performed on receiver-transmitter and Coder-Control, Interrogator Set KY-97( )/TPX in their transit cases, and Case Receiver -Transmitter CY-1324( )/UPX-6 empty.

4.7.1 Preproduction test. - The equipment, as packed for field transport, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be approximately 40° F. above the the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water at approximately its initial temperature, or the temperature of the water shall be maintained constant by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. Theree shall be no evidence of leakage.

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4.7.2 Air-seal test.- The equipment, as packed for field transport, shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment inclosure when no transit case is provided. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily.

4.8 Temperature test for equipment.- The equipment shall be subjected to the temperature cycle shown on Drawing SC-D-19999 with exception as specified herein. Steps 2A, 2B, and 3 shall be limited to 125° F (+6, -0°F.). Measurements shall be taken as follows:

Measurement	Reference paragraph	Step	Test results
Receiver performance	4.16	4,5,8,9	Full specification performance
Transmittter and coder-control	4.17	4,5,8,9	Full specification performance
Coder-Control (KY-99( )/UPX)	4.18	4,5,8,9	Full specification performance

4.9 Moisture-resistance test.-

4.9.1 Test conditions.-

a. Do not remove equipment from the humidity chamber for measurements.
b. Start measurements not more than 10 minutes after power is applied to the components. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.
c. Do not open sealed cases for the purposes of making measurements. Instead, bring out leads through a hole in the case and seal the hole against entrance of moisture.
d. For sealed components, perform the last two cycles with the cases removed.

4.9.2 Test procedure.- The equipment will be tested as follows:

a. Dry at 125 ±5°F for 24 hours.
b. Condition at 77 ±5°F and 40 to 50 percent relative humidity for 24 hours.

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c. Make measurements as designated in 4.9.3 and readjust or realign if necessary to meet specification requirements. Performance shall be full specification requirements.

d. Subject to continuous cycling for five (5) 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Drawing SC-D-15914 with exceptions herein. The test of 4.19 will be made during the periods shown on the drawing and shall comply with 4.9.3. The temperature limit shall be 131° F instead of 149° F as shown in Drawing SC-D-15914.

e. After cycling has been completed, condition the components for 24 hours at 77 ±5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the components. No repair or replacement of parts will be made. After adjustments, the components shall meet full specification requirements for those measurements listed in 4.9.3.

4.9.3 Performance.- Measurements made during tests, and required performance during cycling, shall be as follows:

	Measurement	Ref. Par.	Performance During Cycling
(a)	Sensitivity	4.16.1	Shall comply with 3.9.5.1.2
(b)	Power output	4.17.1	shall comply with 3.9.5.2.1

4.9.4 Failure.- If any component fails to meet the performance listed in 4.9.3 during cycling, it will be realigned or readjusted once. If the component then fails to meet 4.9.3 or fails subsequently during cycling, it does not pass the test.

4.10 "Normal" testing conditions.- The following conditions of testing shall be considered "normal" for the purpose of this specification:

(a)	Temperature	-Normal atmospheric
(b)	Pressure	-Normal atmospheric
(c)	Humidity	-Normal atmospheric
(d)	Line Voltage	-117 volts ±10 percent
(e)	Antenna terminal matched load	-52 ohms with SWR (standing wave ratio) not greater than ±0.2 db
(f)	Equipment position	-Vertical and stationary
(g)	Operating period	-Begins after a 10 minute warm-up
(h)	Vacuum tubes	-Random choice
(i)	Power supply	-Frequency constant at 400 cps or 60 cps as applicable

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4.10.1 Transmitter and receiver measurements.- Unless otherwise stated all measurements of transmitter and receiver performance shall be made as specified in the following paragraphs: Except in tests where greater accuracy is required the standard for these measurements shall be considered as properly adjusted and calibrated Radar Test Set AN/UPM-4( ), or an approved equal.

4.10.1.1 Terminal for radio frequency measurements.- When radio-frequency measurements are being conducted, the measuring equipment shall be connected to the antenna terminal on the equipment.

4.10.1.2 Signal for receiver input measurements.- Signals supplied to the receiver unit when measurements are being conducted shall be derived from a 52-ohm signal generator.

4.10.1.3 Receiver output measurements.- The voltage developed across a 75-ohm resistance load at the output terminal of the receiver shall be considered as the normal receiver output. No attenuation shall be introduced between the output amplifier of the receiver and the output terminal during measurements.

4.11 Variation in supply line volatage.- The line voltage shall be varied from 10 percent below to 10 percent above the normal value during a period of 5 minutes without permanent damage to the equipment.

4.12 Change of tubes.- In order to determine that the equipment will perform within the limits of this specification with a random selection of vacuum tubes, the following test shall be performed on the preproduction sample, the tenth and twentieth units of production.

4.12.1 Twenty complete sets of tubes, composed of types representing all the Signal Corps approved manufacturers selected at random and preferably from different production runs of each manufacturer, shall be substituted in the equipment one complete set at a time. The equipment shall operate within limits of this specification with each set of tubes. Any tubes responsible for failure of the equipment shall be set aside for tests to determine compliance with JAN limits.

4.13 Microphonic tubes.- To determine if any tube is microphonic, the equipment shall be connected with input signal and output display as for measurement of receiver sensitivity, and with the output level just below saturation. Each tube of the equipment with the exception of the transmitter-oscillator tube, shall be tapped with a small rubber hammer. If the tapping of any tube causes substantial distortion of the output, that tube shall be regarded as microphonic ans shall be rejected.

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4.14 Twenty-four hour performance tests. - The receiver-transmitter and the Coder-Control, Interrogator Set KY-97( )/TPX shall be given 24-hour run check in accordance with the procedure described in the following sub-paragraphs:

4.14.1 The voltages at the various terminals shall be checked.

4.14.2 The i-f circuits shall be aligned.

4.14.3 The pulse length and power output at 1030 mc shall be checked; the power output may be checked at the r-f probe. The sensitivity at 1090 mc shall be checked.

4.14.4 The equipment shall be adjusted to transmit on 1030 megacycles and receive at 1090 megacycles.

4.14.5 The equipment shall be operated continuously under full load for an interval of 24 hours in the same manner as it would be operated while in field service. During the interval the transmitter shall be triggered continuously at the rate of approximately 300 pulses per second. The operation of the equipment shall be observed at reasonable intervals to determine if it is operating properly, and improper operation shall be investigated. In the event that the improper operation is caused by the failure of minor component parts such as resistors, small condensers or tubes, or by misadjustment, the equipment shall be repaired and the operation continued for a time interval such that the time of operation prior to the failure plus the time of operation following failure equals 24 hours. In the event that defective operation is caused by the failure of a major component part such as a power transformer, the equipment shall be considered as failing to pass this test. A complete record of all observations made during this test shall be forwarded to the Government laboratory concerned.

4.15 Corona test. - The receiver-transmitter and Coder-Control, Interrogator Set KY-97( )/TPX shall be operated at repetition rate such that the receiver output, as observed on an oscilloscope, is synchronized with the power supply (i.e., repetition rate of 400 pulses per second). The gain of the receiver shall be turned to maximum and the oscilloscope display of the output observed. If spurious signals, manifesting corona, are observed in the display, the equipment fails this test.

4.16 Performance tests for receiver circuit. -After the overall 24-hour test as specified in 4.14 is completed, the receiver circuit shall be subjected to the performance tests specified in the following subparagraphs:

4.16.1 Sensitivity. - The receiver's sensitivity shall be measured at 1090 mc and any other frequency selected at the discretion of the Government inspector. The measured sensitivity shall comply with 3.9.5.1.2.

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4.16.2 Image ratio. - The receiver's image ratio shall be determined at 1090 mc and any other frequency selected at the discretion of the Government Inspector, and shall be no worse than 70 db.

4.16.3 Bandwidth. - The receiver's overall bandwidth shall be measured at 1090 mc and any other frequency selected at the discretion of the Government inspector, and shall comply with 3.9.5.1.3.

4.16.4 Output level. - The receiver's dynamic output range shall be measured at one frequency selected at the option of the Government inspector, and shall comply with 3.9.5.1.12.

4.16.5 Video output. - The video output of the receiver shall be measured at 1090 mc and shall comply with 3.9.5.1.12.

4.16.6 I-F gain control. - With an impressed signal of 30 db below 1 volt at 1090 mc, the i-f gain control shall be tested. The gain expressed as a function of the angle of rotation of the knob, shall increase approximately logarithmically over a range of at least 55 db.

4.16.7 Gain-time control and gate. - The gain-time control and gate facilities of the receiver circuit shall be measured at 1090 mc to determine compliance with 3.9.5.1.6. In particular, the following features shall be observed with mode switch set to Mode 1:

a. Receiver gain, when receiver-transmitter is not triggered.
b. Rise of receiver gain during 10 microseconds after equipment is triggered.
c. Rise of receiver gain after elapse of 10 microseconds following triggering of equipment. Gain rise for Conditions 1 and 2 shall be measured with gain attenuation setting #1. (See 3.9.5.1.6.1.1 and 3.9.5.1.6.1.2)
d. Duration at normal gain as specified in 3.9.5.1.6.2.

4.16.8 Anti-jamming. - The anti-jamming facilities of the receiver shall be tested at 1090 mc and shall comply with 3.9.5.1.8.

4.16.9 Fidelity and recovery. - The fidelity and recovery characteristics of the receiver shall be determined at 1090 mc as outlined in 3.9.5.1.9.

4.16.10 Receiver delay. - The receiver delay shall be checked at 1090 mc and shall comply with 3.9.5.1.10.

4.16.11 Intermediate -frequency pickup. - Compliance with 3.9.5.1.11 shall be checked.

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4.17 Performance tests for transmitter and coder-Control, Interrogator Set KY-97( )/TPX circuits.- After the overall 24-hour test as specified in 4.14 is completed, the transmitter and coder circuits shall be subjected to the performance tests specified in the following subparagraphs:

4.17.1 Power output.- The power output of the transmitter shall be measured at 1030 mc and shall comply with 3.9.5.2.1.

4.17.2 R-F pulse shape.- The duration and shaping of the transmitter pulse trains shall be determined at 1030 mc and at any other frequencies selected at the discretion of the Government inspector. The duration shall be between 0.7 and 1.2 microseconds. The shaping shall comply with 3.9.5.2.7.

4.17.3 Transmitter delay.- Compliance with 3.9.5.2.8 shall be checked at 1030 mc.

4.17.4 Trigger out pulse.- The amplitude and shape of the trigger out pulse shall be determined at 1030 mc and shall comply with 3.9.5.3.1.

4.17.5 Operation.- The operation of the transmitter and coder shall be tested at 1030 mc with input synchronizing pulses as follows:

	Synch pulse 1	Synch pulse 2
Amplitude	5 volts	20 volts
Repetition rate	4100 *pps*	100 cycles

4.17.6 Pulse spacings for the three modes.- The pulse spacing shall be checked for compliance with 3.10.2.2.6.

4.17.7 Chop rate.- The chop rate shall be checked for compliance with 3.10.2.2.8.

4.18 Performance of Coder-Control, Interrogator Set KY-99( )/UPX.- Tests shall be performed on this equipment using a Receiver-Transmitter, Radio RT-264( )/UPX-6 or a similated test set up as called for in bid request and contract. The procedure for tests shall be similar to those of the Coder-Control KY-97( )/TPX (see 4.17) but meeting requirements applicable to Coder-Control KY-99( )/UPX. The following tests shall be performed:

	Ref. Par.
a. Twenty-four hour performance	3.11.2
b. Triggering	3.11.2.1
c. Self trigger	3.11.2.2
d. Control circuit output	3.11.2.5
e. Interrogation mode change	3.11.2.6
f. Challenge	3.11.2.7

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4.19 Visual and mechanical inspection.- Parts and equipment shall be inspected for workmanship, mechanical fit, loose nuts, and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connection, ground connections, welds, finishes, etc., shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government inspector as major or minor in accordance with the definitions of Standard MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment.- Components for Interrogator Sets shall be packaged in accordance with Method III of Specification MIL-P-116.

5.1.2 For export shipment.- Preserve and package components for Interrogator Sets in accordance with procedures specified for the designated methods as prescribed in Specification MIL-P-116, and further described in the following paragraphs.

5.1.2.1 Technical literature.- Package each technical manual Method 1C-3, as follows Inclose each technical manual within a close-fitting Type II Class b bag, conforming to Specification JAN-P-117. Properly seal closure.

5.1.2.2 Receiver-Transmitter, Radio RT-211( )/TPX in Case, Receiver-Transmitter CY-944( )/TPX; Coder-Control, Interrogator Set KY-99( )/UPX; Case, Standardized components Electrical CY-951( )/TPX; Case, Receiver-Transmitter CY-1324( )/UPX-6.- Package each component Method 1C-5 as follows: As required cushion each component on all surfaces with cells or pads or both, fabricated of corrugated fiberboard, conforming to Specification 19-10 and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned component within a close-fitting suitable style and grade fiberboard box conforming to Specification JAN-P-108. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape, conforming to Specification JAN-P-127.

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5.1.2.3 Coder-Control, Interrogator Set KY-97( )/TPX. - Package each unit Method IIb, as follows: Cushion each unit on all surfaces with cells or pads or both, fabricated of corrugated fiberboard, conforming to Specification 19-10 and designed to absorb the shock of impact normally encountered in handling and transit, and to square-up the unit. Place each cushioned unit together with the required amount of desiccant as prescribed in Specification MIL-P-116, within a close-fitting suitable style and grade corrugated fiberboard box. Seal entire closure with gummed Kraft tape conforming to Specification UU-T-111. Blunt allcorners of the box. Place the boxed unit within a water-vapor-proof barrier, qualified for class 1 application in accordance with requirements of Specification MIL-B-131. Heat seal closure. Place the water-vapor-proofed unit within a close-fitting suitable style and grade fiberboard box conforming to Specification JAN-P-108. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape, conforming to Specification JAN-P-127.

5.2 Packing. -

5.2.1 For domestic shipment. - Components for Interrogator Sets, packaged as specified in 5.1.1, shall be packed in substantial containers of the type, size, and kind commonly used for the purpose, conforming to Specification LLL-B-631, NN-B-591, NN-B-601, or NN-B-631, in such a manner as to insure acceptance at and safe delivery to the designated point by common or other carrier at the lowest rate.

5.2.2 For export shipment. - The components for Interrogator Sets shall be individually packed in a wood cleated plywood nailed wood or wire bound wood box conforming to the requirements of Specifications JAN-P-105, JAN-P-106, and MIL-B-107 respectively. Fabricate the boxes to fit contents snugly. Close the shipping container as specified in the appendix of the applicable box specification.

5.2.2.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.3 Marking for shipment. - Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD -129.

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6. NOTES

6.1 Intended use. - The components covered by this specification provide I. F. F. for various radar sets.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification
b. The component required and whether cases are required.
c. Whether domestic or overseas shipment is required.
d. Samples required. (See 3.2)
e. Marking and shipping of samples.
f. Place of final inspection. (See 4.1)
g. Acceptable quality level (See 4.1.1, 4.1.2.1 and table III).
h. Facsimiles required (See 3.6)
i. Technical literature required. (See 3.14)
j. Quantity of tools and running spare parts required. (See 3.14)

6.3 Contractual requirements. - It is recommended that the following be included in the contract.:

6.3.1 Additional testing. - It should be understood that additional inspection and nondestructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents. The Government inspector, therefore, may withdraw materials parts, subassemblies, or components temporarily from production for such inspection and testing, performed either at a Government laboratory or the contractor's plant.

6.3.2 Waiver of preproduction samples. - When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that the specified samples have been submitted and approved on the current contract; Production of the item under the new order will be concurrent with, or follow without interruption, production on the current order; and current production is in strict compliance with specific requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

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6.4 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example RT-211/TPX. As soon as possible after the award of the contract, the contractor should apply to the contracting officer for such information. (See 1.1.)

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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